Mekanik.py Formula Sheet

A Python-Based Mechanical Engineering Formula Sheet

Analytical formulas and computational interfaces

Library name:	`mekanik.py`
Domain:	Mechanical engineering
Implementation:	Python
Version:	0.2

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Introduction

This document presents `mekanik.py`, a compact Python-based reference library for classical mechanical engineering calculations. Its purpose is to connect analytical formulas commonly found in engineering handbooks with their direct computational use in engineering analysis, design, and verification.
The content is organized as a formula sheet structured by topic. Each analytical expression is paired with a corresponding Python function, creating a clear one-to-one correspondence between theory and implementation. This structure emphasizes transparency and traceability, and all formulas are written using standard mechanical engineering notation and units.
The current version covers tolerances according to ISO 286, bolted joints, springs, interference fits, and basic models for brakes and wear. The focus is on clarity and correctness rather than numerical optimization or advanced numerical methods. As such, the library is well suited for preliminary design calculations, parametric studies, educational use, and the verification of numerical results.
The document is designed to be extensible. Additional chapters can be added using the same structure: a clear analytical formulation, a consistent set of symbols, and a minimal, explicit Python interface. In this way, `mekanik.py` serves both as a practical engineering tool and as a concise, self-contained mechanical engineering reference.

Contents

1 Tolerances

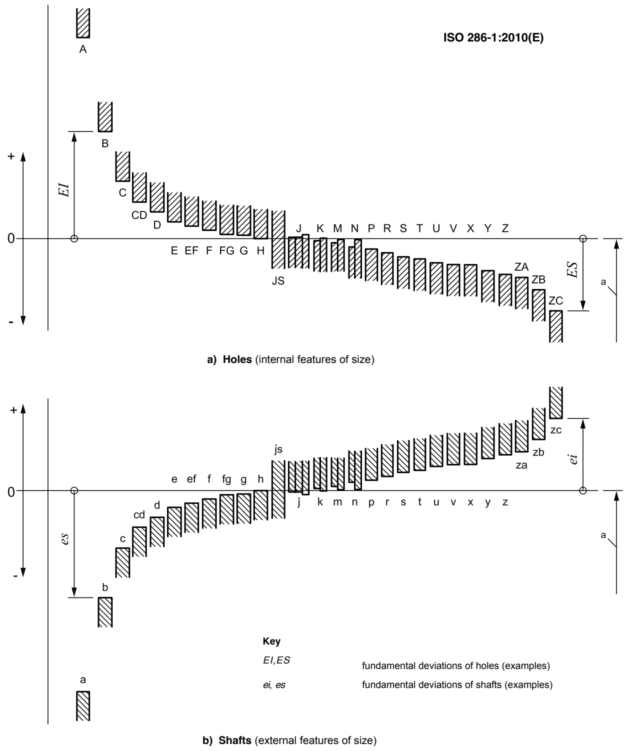


Figure 1: Schematic representation of tolerance positions

1.1 Tolerance definitions

$$\textbf{Tolerance (width): } T = U - L \tag{1}$$

$$\textbf{Deviations from nominal (hole): } ES = U - D, \qquad EI = L - D \tag{2}$$

$$\textbf{Deviations from nominal (shaft): } es = U - D, \qquad ei = L - D \tag{3}$$

1.2 IT grades

Note: Internally, IT values are handled in μm and converted to mm when combined with nominal dimensions.

$$\textbf{IT grade tolerance value: } IT = IT(D,\ g) \tag{4}$$

```
IT = it_um(nominal_mm, grade)   # IT in micrometers
```

1.3 Fundamental deviations

$$\textbf{Tolerance limits from designation (e.g. H7, h6, JS7): } (L,\ U) = \mathrm{iso\_tol}(D,\ \mathrm{tol}) \tag{5}$$

```
(lower, upper) = iso_tol(nominal_mm, tol, mode="abs")  # absolute limits [mm]
(low_dev, high_dev) = iso_tol(nominal_mm, tol, mode="dev")  # deviations [mm]

(lower, upper) = iso_tol(10, 'H7', "abs", "mm")
```

$$\textbf{Symmetric zone (JS/js): } \Delta_{\min} = -\frac{IT}{2}, \qquad \Delta_{\max} = +\frac{IT}{2} \tag{6}$$

$$\textbf{Generic construction from one deviation value + IT: } \begin{cases} L = D + \Delta_{\min} \\ U = D + \Delta_{\max} \end{cases} \tag{7}$$

1.4 Fits

$$\textbf{Fit clearance range (hole - shaft): } c_{\min} = L_h - U_s, \qquad c_{\max} = U_h - L_s \tag{8}$$

$$\textbf{Overlap of tolerance intervals (size space): } \ell_{\text{overlap}} = \max\Big(0,\ \min(U_h, U_s) - \max(L_h, L_s)\Big) \tag{9}$$

```
fit = fit_check(nominal_mm, hole_tol, shaft_tol)
# hole_tol example: "H7"
# shaft_tol example: "h6"
```

The returned dictionary from `fit_check` contains:

```
fit["hole_limits"]    # (Lh, Uh)
fit["shaft_limits"]   # (Ls, Us)
fit["clearance"]      # (c_min, c_max)
fit["overlap"]        # overlap length
fit["fit_type"]       # "clearance" / "transition" / "interference"
```

Table 1: Symbols used in ISO 286 tolerance calculations

Symbol	Description	Unit
D	Nominal size	mm
L	Lower limit size	mm
U	Upper limit size	mm
T	Tolerance width $(U - L)$	mm
ES	Upper deviation (hole) $(U - D)$	mm
EI	Lower deviation (hole) $(L - D)$	mm
es	Upper deviation (shaft) $(U - D)$	mm
ei	Lower deviation (shaft) $(L - D)$	mm
IT	IT grade tolerance value	μm or mm
FD	Fundamental deviation (ISO 286)	μm
L_h, U_h	Hole limits	mm
L_s, U_s	Shaft limits	mm
$c_{\min}$	Minimum clearance $(L_h - U_s)$	mm
$c_{\max}$	Maximum clearance $(U_h - L_s)$	mm
ℓ_{overlap}	Interval overlap length	mm

2 Bolts

2.1 Forces

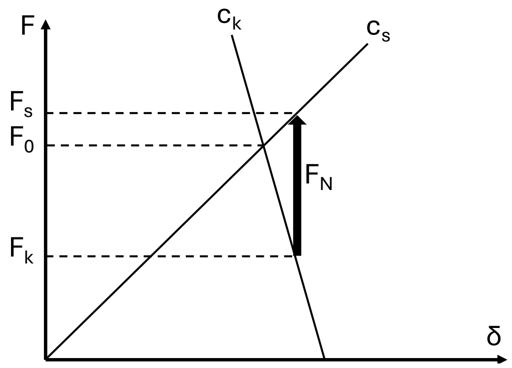


Figure 2: F-delta Diagram

$$\textbf{Screw force: } F_s = F_0 + \frac{c_s}{c_s + c_k} F_N \tag{10}$$

```
Fs = F_s(F0, FN, cs, ck)
```

$$\textbf{Clamping force: } F_k = F_0 - \frac{c_k}{c_s + c_k} F_N \tag{11}$$

```
Fk = F_k(F0, FN, cs, ck)
```

2.2 Torques

$$\textbf{Total tightening torque: } M = F_{\text{ax}} \, (0.16 \, P + 0.58 \, \mu_{\text{thread}} \, d_2 + \mu_{\text{head}} \, r_m) \tag{12}$$

```
M = M_total(Fax, P, d2, r_m, mu_head, mu_thread)
```

$$\textbf{Bearing friction torque: } M_b = \mu_b \, F_b \, r_m \tag{13}$$

```
M_b = M_bearing(mu_b, Fb, r_m)
```

$$\textbf{Thread tightening torque: } M_t = \frac{F_{\text{ax}} \, P \, \tan(\varphi + \rho')}{2\pi \, \tan\varphi} \tag{14}$$

```
M_t = M_tightening(Fax, r_ax, phi, rho, P)
```

$$\textbf{Thread loosening torque: } M_l = -\frac{F_{\text{ax}} \, P \, \tan(\varphi - \rho')}{2\pi \, \tan\varphi} \tag{15}$$

```
M_l = M_loosening(Fax, r_ax, phi, rho, P)
```

Lookup of standard torques for bolts M1.6 to M72; 4.6, 5.8, 8.8, 10.9 and 12.9 3 (16)

```
bolt_torque("M6", "8.8")

--> {'pitch_mm': 1.0, 'area_mm2': 20.1, 'torque_nm': 9.8}
```

2.3 Geometry

$$\textbf{Apparent friction angle: } \rho' = \arctan\left(\frac{\mu}{\cos\alpha}\right) \tag{17}$$

```
rho = rho_apparent(mu, alpha)
```

$$\textbf{Lead angle: } \varphi = \arctan\left(\frac{P}{\pi d_2}\right) \tag{18}$$

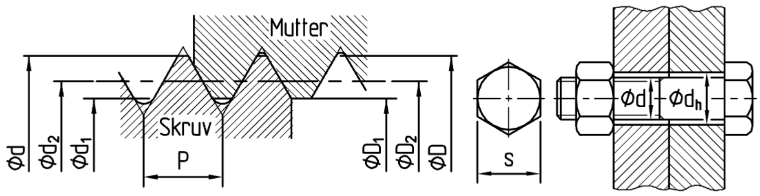
```
phi = phi_lead(P, d2)
```

$$\textbf{Thread stress area: } A_s \approx \frac{\pi}{16}\left(d_1 + d_2\right)^2 \tag{19}$$

```
As = A_s(d1, d2)
```

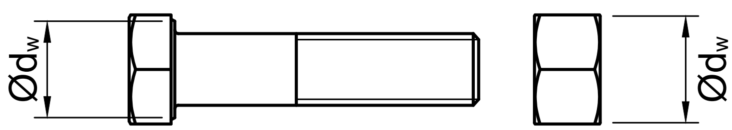
Metric thread dimensions: Lookup of standard ISO metric thread geometry from table 2.3 (20)

```
dims = M_dimensions(d_nom1)

P  = dims["P"]
d  = dims["d"]
d2 = dims["d2"]
d1 = dims["d1"]

dh_fine   = dims["dh_fine"]
dh_medium = dims["dh_medium"]
dh_coarse = dims["dh_coarse"]
```



Thread	P [mm]	d [mm]	d_2 [mm]	d_1 [mm]	$d_{h,\mathrm{f}}$ [mm]	$d_{h,\mathrm{m}}$ [mm]	$d_{h,\mathrm{c}}$ [mm]
M1	0.25	1	0.838	0.729	1.1	1.2	1.3
M1.2	0.25	1.2	1.038	0.929	1.3	1.4	1.5
M1.4	0.30	1.4	1.205	1.075	1.5	1.6	1.8
M1.6	0.35	1.6	1.373	1.221	1.7	1.8	2.0
M1.8	0.35	1.8	1.573	1.421	2.0	2.1	2.2
M2	0.40	2.0	1.740	1.567	2.2	2.4	2.6
M2.5	0.45	2.5	2.208	2.013	2.7	2.9	3.1
M3	0.50	3.0	2.675	2.459	3.2	3.4	3.6
M4	0.70	4.0	3.545	3.242	4.3	4.5	4.8
M5	0.80	5.0	4.480	4.134	5.3	5.5	5.8
M6	1.00	6.0	5.350	4.917	6.4	6.6	7.0
M8	1.25	8.0	7.188	6.647	8.4	9.0	10.0
M10	1.50	10	9.026	8.376	10.5	11.0	12.0
M12	1.75	12	10.863	10.106	13.0	13.5	14.5
M16	2.00	16	14.701	13.835	17.0	17.5	18.5
M20	2.50	20	18.376	17.294	21.0	22.0	24.0
M24	3.00	24	22.051	20.752	25.0	26.0	28.0
M30	3.50	30	27.727	26.211	31.0	33.0	35.0
M36	4.00	36	33.402	31.670	37.0	39.0	42.0
M42	4.50	42	39.077	37.129	43.0	45.0	48.0
M48	5.00	48	44.752	42.587	50.0	52.0	56.0
M56	5.50	56	52.428	50.046	58.0	62.0	66.0
M64	6.00	64	60.103	57.505	66.0	70.0	74.0

Figure 3: Metric thread dimensions. The schematic defines the geometric quantities used for ISO metric threads, and the table lists standard values for common thread sizes.



	M5	**M6**	**M8**	**M10**	**M12**	**M16**
d_w, bolt head [mm]	6.74	8.74	11.47	14.47	16.47	22.00
d_w, nut [mm]	6.90	8.90	11.47	14.60	16.60	22.50
	M20	**M24**	**M30**	**M36**	**M42**	**M48**
d_w, bolt head [mm]	27.70	33.25	42.75	51.11	59.95	69.45
d_w, nut [mm]	27.70	33.25	42.75	51.11	59.95	69.45

Figure 4: Effective contact diameter d_w for bolt head and nut. The schematic defines d_w, and the table lists standard values for common ISO metric threads.

Table 2: Symbols used in bolt calculations

Symbol	Description	Unit
F_s	Bolt force	N
F_k	Clamping force	N
F_0	Preload force	N
F_N	External axial load	N
F_{ax}	Axial bolt force	N
F_b	Bearing force	N
c_s	Bolt stiffness	N/m
c_k	Clamped parts stiffness	N/m
M	Total tightening torque	Nm
M_b	Bearing friction torque	Nm
M_t	Thread tightening torque	Nm
M_l	Thread loosening torque	Nm
P	Thread pitch	m
d_1	Minor diameter	m
d_2	Pitch diameter	m
r_m	Mean bearing / friction radius	m
φ	Thread lead angle	rad
ρ'	Apparent friction angle	rad
α	Thread flank angle	rad
μ	Thread friction coefficient	–
μ_{thread}	Thread friction coefficient	–
μ_{head}	Bearing (under-head) friction coefficient	–
μ_b	Bearing surface friction coefficient	–
A_s	Tensile stress area of thread	m^2

Table 3: Screw Database: Dimensions and Tightening Torques for Property Classes

			Tightening Torque by Property Class (Nm)				
Size	Pitch (mm)	Area (mm^2)	4.6	5.8	8.8	10.9	12.9
M1.6	0.35	1.27	0.065	0.1	0.17	0.24	0.29
M1.8	0.35	1.7	0.096	0.16	0.25	0.36	0.43
M2	0.4	2.07	0.13	0.22	0.35	0.49	0.58
M2.2	0.45	2.48	0.17	0.29	0.46	0.64	0.77
M2.5	0.45	3.39	0.26	0.44	0.7	0.98	1.2
M3	0.5	5.03	0.46	0.77	1.2	1.7	2.1
M3.5	0.6	6.78	0.73	1.2	1.9	2.7	3.3
M4	0.7	8.78	1.1	1.8	3.9	4	4.9
M4.5	0.75	11.3	1.6	2.6	4.1	5.8	7
M5	0.8	14.2	2.2	3.6	5.7	8.1	9.7
M6	1.0	20.1	3.7	6.1	9.8	14	17
M8	1.25	36.6	8.9	15	24	33	40
M10	1.5	58.0	17	29	47	65	79
M12	1.75	84.3	30	51	81	114	136
M14	2.0	115.0	48	80	128	181	217
M16	2.0	157.0	74	123	197	277	333
M18	2.5	192.0	103	172	275	386	463
M20	2.5	245.0	144	240	385	541	649
M22	2.5	303.0	194	324	518	728	874
M24	3.0	353.0	249	416	665	935	1120
M27	3.0	459.0	360	600	961	1350	1620
M30	3.5	561.0	492	819	1310	1840	2210
M33	3.5	694.0	663	1100	1770	2480	2980
M36	4.0	817.0	855	1420	2280	3210	3850
M39	4.0	976.0	1100	1830	2930	4120	4940
M42	4.5	1121.0	1360	2270	3540	5110	6140
M45	4.5	1306.0	1690	2820	4510	6340	7610
M48	5.0	1473.0	2040	3400	5450	7660	9190
M52	5.0	1758.0	2620	4370	6990	9830	11800
M56	5.5	2030.0	3270	5440	8710	12200	14700
M60	5.5	2362.0	4050	6750	10800	15200	18200
M64	6.0	2676.0	4900	8170	13100	18400	22000
M68	6.0	3055.0	5910	9860	15800	22200	26600
M72	6.0	3460.0	7060	11800	18800	26500	31800

3 Springs

3.1 Spring stiffness

$$\textbf{Axial spring constant (helical spring):}\ c_a = \frac{G\,d^4}{8\,n\,D^3} \tag{21}$$

```
ca = spring_constant_axial(G, d, D, n)
```

$$\textbf{Torsional spring constant:}\ c_v = \frac{E\,d^4}{64\,n\,D} \tag{22}$$

```
cv = spring_constant_torsional(E, d, D, n)
```

3.2 Stresses

$$\textbf{Torsional shear stress:}\ \tau_v = \frac{8\,F\,D}{\pi\,d^3} \tag{23}$$

```
tau_v = shear_stress_torsion(F, D, d)
```

$$\textbf{Effective shear stress (with correction):}\ \tau_e = k\,\frac{8\,F\,D}{\pi\,d^3} \tag{24}$$

```
tau_e = shear_stress_effective(F, D, d, k)
```

$$\textbf{Wahl correction factor:}\ k = \frac{4c-1}{4c-4} + \frac{0.615}{c}, \qquad c = \frac{D}{d} \tag{25}$$

```
k = wahl_factor(D, d)
```

3.3 Geometry and stability

$$\textbf{Minimum free spring length:}\ l_0 = 1.25\,(n+1)\,d + \delta \tag{26}$$

```
l0 = free_length(n, d, delta)
```

$$\textbf{Buckling criterion for compression springs:}\ l_0 < 2.6\,D \tag{27}$$

```
stable = check_buckling(l0, D)
```

3.4 Energy and work

$$\textbf{External work (linear spring):}\ W = \frac{1}{2}\,F\,\delta \tag{28}$$

```
W = external_work_linear(F, delta)
```

$$\textbf{External work (rotational spring):}\ W = \frac{1}{2}\,M\,\varphi \tag{29}$$

```
W = external_work_rotational(M, phi)
```

$$\textbf{Maximum storable energy (normal stress):}\ W_{\max} = V\,\frac{\sigma_{\max}^2}{2\,E} \tag{30}$$

```
Wmax = max_energy_density_normal(sigma_max, E, V)
```

$$\textbf{Maximum storable energy (shear stress):}\ W_{\max} = V\,\frac{\tau_{\max}^2}{2\,G} \tag{31}$$

```
Wmax = max_energy_density_shear(tau_max, G, V)
```

$$\textbf{Utilization factor:}\ \eta = \frac{W_{\text{actual}}}{W_{\max}}, \qquad 0 < \eta < 1 \tag{32}$$

```
eta = utilization_factor(W_actual, W_max)
```

Table 4: Symbols for spring formulas

Symbol	Description	Unit
E	Young's modulus	Pa
G	Shear modulus	Pa
d	Wire diameter	m
D	Mean coil diameter	m
n	Number of active coils	–
c	Spring index, $c = D/d$	–
c_a	Axial spring constant (stiffness)	N/m
c_v	Torsional (rotational) spring constant	N·m/rad
F	Spring force (axial load)	N
M	Torque	N·m
δ	Axial deflection	m
φ	Angular deflection	rad
τ_v	Torsional shear stress (uncorrected)	Pa
τ_e	Effective shear stress (corrected)	Pa
k	Correction factor (Wahl factor)	–
l_0	Free spring length	m
W	External work / stored elastic energy	J
V	Loaded volume	m^3
$\sigma_{\max}$	Maximum normal stress	Pa
$\tau_{\max}$	Maximum shear stress	Pa
$W_{\max}$	Maximum storable elastic energy	J
W_{actual}	Actual stored elastic energy	J
η	Utilization factor	–

4 Interference fits

4.1 Friction forces

$$\textbf{Frictional force at macro-sliding: } F = \mu\, \pi\, d\, L\, p = \mu\, N \tag{33}$$

```
F = friction_force(mu, p=p, d=d, L=L)
# or
F = friction_force(mu, N=N)
```

$$\textbf{Resultant friction force: } F = \sqrt{F_{\text{ax}}^2 + F_{\text{per}}^2} \tag{34}$$

```
F = friction_force_resultant(F_axial, F_peripheral)
```

$$\textbf{Transmitted torque: } M = \frac{d}{2}\, F_{\text{per}} \tag{35}$$

```
M = torque_transmitted(F_peripheral, d)
```

4.2 Geometry relations

$$\textbf{Hub geometry ratio: } \kappa = \frac{d}{D} \tag{36}$$

```
k = kappa(d, D)
```

$$\textbf{Shaft geometry ratio (hollow shaft): } \kappa_0 = \frac{d_0}{d} \tag{37}$$

```
k0 = kappa0(d0, d)
```

4.3 Interference and contact pressure

$$\textbf{Interface pressure from interference: } p = \frac{\Delta}{d\left[\frac{1}{E_{\text{hub}}}\left(\frac{1+\kappa^2}{1-\kappa^2} + \nu_{hub}\right) + \frac{1}{E_{\text{shaft}}}\left(\frac{1+\kappa_0^2}{1-\kappa_0^2} + \nu_{shaft}\right)\right]} \tag{38}$$

```
p = interference_pressure(delta, d, d0, D, E_hub, E_shaft, nu_hub, nu_shaft)
```

$$\textbf{Required diametral interference: } \Delta = p\, d\left[\frac{1}{E_{\text{hub}}}\left(\frac{1+\kappa^2}{1-\kappa^2} + \nu_{hub}\right) + \frac{1}{E_{\text{shaft}}}\left(\frac{1+\kappa_0^2}{1-\kappa_0^2} + \nu_{shaft}\right)\right] \tag{39}$$

```
delta = required_interference(p, d, d0, D, E_hub, E_shaft, nu_hub, nu_shaft)
```

4.4 Stresses after press fit

$$\textbf{Maximum equivalent stress (hub, Tresca): } \sigma_{e,\max}^{T} = p\,\frac{2\kappa^2}{1-\kappa^2} \tag{40}$$

$$\textbf{Maximum equivalent stress (hub, von Mises): } \sigma_{e,\max}^{vM} = p\,\frac{4\kappa^2}{3(1-\kappa^2)}(1+\kappa^2) \tag{41}$$

```
sigma = stresses_after_fit(p, k, material="hub", criterion="Tresca"  # or "vonMises")
```

$$\textbf{Maximum equivalent stress (hollow shaft): } \sigma_{e,\max} = p\,\frac{2\kappa_0^2}{1-\kappa_0^2} \tag{42}$$

```
sigma = stresses_after_fit(p, k, k0, material="hollow_shaft")
```

$$\textbf{Maximum equivalent stress (solid shaft): } \sigma_{e,\max} = p \tag{43}$$

```
sigma = stresses_after_fit(p, k, material="solid_shaft")
```

4.5 Thermal assembly

$$\textbf{Required temperature difference for assembly: } \Delta T = \frac{\Delta}{\alpha\, d} \tag{44}$$

```
dT = required_temperature_difference(delta, d, alpha)
```

Table 5: Symbols for interference fit formulas

Symbol	**Description**	**Unit**
μ	Coefficient of friction	–
p	Interface (contact) pressure	Pa
d	Interface diameter / fit diameter	m
D	Hub outer diameter	m
d_0	Shaft inner diameter (hollow shaft)	m
L	Contact length	m
κ	Hub geometry ratio, $\kappa = d/D$	–
κ_0	Shaft geometry ratio, $\kappa_0 = d_0/d$	–
Δ	Diametral interference	m
F	Frictional (tangential) force	N
F_{ax}	Axial friction force component	N
F_{per}	Peripheral friction force component	N
N	Normal force at the interface, $N = \pi dLp$	N
M	Transmitted torque	N·m
E_{hub}	Young's modulus (hub)	Pa
E_{shaft}	Young's modulus (shaft)	Pa
ν_{hub}	Poisson's ratio (hub)	–
ν_{shaft}	Poisson's ratio (shaft)	–
$\sigma_{e,\text{max}}$	Maximum equivalent (effective) stress after fit	Pa
$\sigma_{e,\text{max}}^{T}$	Maximum equivalent stress (Tresca)	Pa
$\sigma_{e,\text{max}}^{vM}$	Maximum equivalent stress (von Mises)	Pa
α	Linear thermal expansion coefficient	1/K
ΔT	Required temperature difference for assembly	K

5 Brakes

5.1 Wear

$$\textbf{Archard's wear law: } Q = \frac{K\,W\,L}{H} \tag{45}$$

```
Q = archard_wear_law(K, W, L, H)
```

where Q is the total volume of wear debris, K is a dimensionless wear coefficient, W is the normal load, L is the sliding distance, and H is the hardness of the softer contacting material.

5.2 Disc brakes

$$\textbf{Disc brake torque (new pads, uniform pressure): } M = \frac{2}{3}\,\mu\,F_E\frac{R_o^3 - R_i^3}{R_o^2 - R_i^2} \tag{46}$$

```
M = disc_brake_new_pads(mu, F_E, R_o, R_i)
```

$$\textbf{Disc brake torque (worn pads, uniform wear): } M = \mu\,F_W\,\frac{R_o + R_i}{2} \tag{47}$$

```
M = disc_brake_worn_pads(mu, F_W, R_o, R_i)
```

Table 6: Symbols for brake and wear formulas

Symbol	Description	Unit
Q	Total wear volume (wear debris produced)	m^3
K	Wear coefficient (dimensionless)	–
W	Normal load	N
L	Sliding distance	m
H	Hardness of the softer contacting material	Pa
μ	Coefficient of friction	–
M	Brake torque (per pad)	N·m
F_E	Normal force per pad (new pads, uniform pressure)	N
F_W	Normal force per pad (worn pads, uniform wear)	N
R_o	Outer radius of pad contact area	m
R_i	Inner radius of pad contact area	m

6 Beams in bending

The `BeamCase` base class provides a unified computational interface for multi-variable batch simulation, spatial boundary handling, numerical differentiation, and state optimization. All analytical and numerical methods natively support vectorized execution across multi-element design spaces.

6.1 Analytical and Numerical Evaluation Endpoints

6.1.1 Elastic Deflection Curve

The vertical downward displacement profile $\Delta(x)$ evaluates the physical sag or upward bow of the beam's elastic curve along its global continuous span coordinate x. To safeguard physical validity during automated batch design space exploration, a boundary tracking mask enforces the spatial constraints:

$$\Delta(x) = \begin{cases} v(x) & \text{when } 0 \leq x \leq \mathcal{L} \\ \text{NaN} & \text{when } x < 0 \text{ or } x > \mathcal{L} \end{cases} \tag{48}$$

```
# Evaluate the continuous downward elastic deflection curve profile
delta_x = beam.deflection(x)
```

6.1.2 Internal Shear Force distribution

The internal shear force profile $V(x)$ tracks the operational transverse slicing forces across the beam's cross-sectional zones, corresponding to the spatial derivative of the internal bending moment distribution.

$$V(x) = \frac{dM(x)}{dx} \tag{49}$$

```
# Fully vectorized internal shear force calculation field
V_x = beam.shear_force(x)
```

6.1.3 Internal Bending Moment distribution

The internal bending moment profile $M(x)$ captures the structural flexural fields across the coordinate plane, serving as the basis for downstream stress and deformation tracking.

$$M(x) = \int V(x)\, dx \tag{50}$$

```
# Fully vectorized internal bending moment calculation field
M_x = beam.bending_moment(x)
```

6.1.4 Beam Slope (Rotation)

The elastic slope curve $\theta(x)$ represents the first spatial derivative of the continuous deflection profile. Where an analytical expression is not explicitly isolated, the library approximates the local angular rotation using a central finite-difference scheme:

$$\theta(x) = \frac{dv(x)}{dx} \approx \frac{v(x + \Delta x) - v(x - \Delta x)}{2\Delta x} \tag{51}$$

Boundary nodes at the outer limits ($x = 0$ and $x = \mathcal{L}$) automatically utilize forward or backward single-sided finite difference partitions to prevent out-of-bounds array tracking.

```
# Calculate numerical beam slope (rotation in radians) at position x
theta_x = beam.slope(x, dx=1e-5)
```

6.1.5 Beam Curvature

The mechanical curvature profile field $\kappa(x)$ defines the localized rate of change of the slope angle, relating the internal bending moment directly to the section's total flexural rigidity:

$$\kappa(x) = \frac{d\theta(x)}{dx} = \frac{M(x)}{EI} \tag{52}$$

```
# Compute synchronized geometric curvature array field
kappa_x = beam.curvature(x)
```

6.1.6 Bending Stress

The normal bending stress $\sigma_b(x)$ developed across the extreme fibers at a distance $y_{\max}$ from the cross-sectional neutral axis is evaluated via the elastic flexure formula:

$$\sigma_b(x) = \frac{|M(x)| \cdot y_{\max}}{I} \tag{53}$$

```
# Calculate extreme fiber normal bending stress distribution at point x
sigma_x = beam.bending_stress(x, y_max)
```

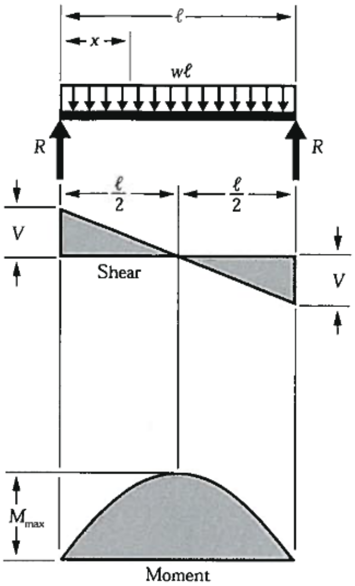
6.2 Global Optimization and Extremum Detection

To locate critical limit-state design thresholds without manual vector searching, the library implements a high-density matrix solver sweep over a spatial grid of $\mathcal{N}$ points across the full length of the beam profile ($\mathcal{L}$).

$$y_{\text{target}} = \max_{x \in [0,\mathcal{L}]} |f_{\text{target}}(x)| \tag{54}$$

Where f_{target} represents the continuous distribution field array for deflection (v), slope (θ), bending moment (M), or shear force (V). The tracking engine scales automatically across 2D batch spaces, isolating column-wise peak design variants instantly.

```
# Query peak structural design limit constraints
v_max     = beam.solve_for("v_max", grid_points=1000)      # Max Deflection
theta_max = beam.solve_for("theta_max", grid_points=1000)  # Max Rotation
M_max     = beam.solve_for("M_max", grid_points=1000)      # Max Bending Moment
V_max     = beam.solve_for("V_max", grid_points=1000)      # Max Shear Force
```



$$R = V \ldots\ldots\ldots\ldots\ldots\cdots = \frac{w\ell}{2}$$

$$V_x \ldots\ldots\ldots\ldots\ldots\cdots = w\left(\frac{\ell}{2} - x\right)$$

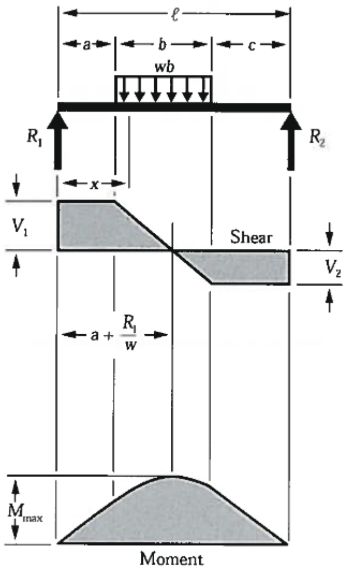
$$M_{\max} \text{ (at center)} \ldots\ldots\ldots\ldots\ldots\cdots = \frac{w\ell^2}{8}$$

$$M_x \ldots\ldots\ldots\ldots\ldots\cdots = \frac{wx}{2}(\ell - x)$$

$$\Delta_{\max} \text{ (at center)} \ldots\ldots\ldots\ldots\ldots\cdots = \frac{5w\ell^4}{384EI}$$

$$\Delta_x \ldots\ldots\ldots\ldots\ldots\cdots = \frac{wx}{24EI}(\ell^3 - 2\ell x^2 + x^3)$$

Figure 5: Case 1



$$R_1 = V_1 \text{ (max when } a < c) \ldots\ldots\cdots = \frac{wb}{2\ell}(2c + b)$$

$$R_2 = V_2 \text{ (max when } a > c) \ldots\ldots\cdots = \frac{wb}{2\ell}(2a + b)$$

$$V_x \text{ (when } x > a \text{ and } < (a + b)) \ldots\cdots = R_1 - w(x - a)$$

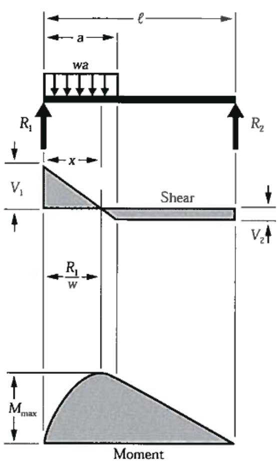
$$M_{\max}\left(\text{at } x = a + \frac{R_1}{w}\right) \ldots\ldots\cdots = R_1\left(a + \frac{R_1}{2w}\right)$$

$$M_x \text{ (when } x < a) \ldots\ldots\ldots\ldots\cdots = R_1 x$$

$$M_x \text{ (when } x > a \text{ and } < (a + b)) \ldots\cdots = R_1 x - \frac{w}{2}(x - a)^2$$

$$M_x \text{ (when } x > (a + b)) \ldots\ldots\cdots = R_2(\ell - x)$$

Figure 6: Case 2



$$R_1 = V_1 \ldots\ldots\ldots\cdots = \frac{wa}{2\ell}(2\ell - a)$$

$$R_2 = V_2 \ldots\ldots\ldots\cdots = \frac{wa^2}{2\ell}$$

$$V_x \text{ (when } x < a) \ldots\ldots\ldots\cdots = R_1 - wx$$

$$M_{\max}\left(\text{at } x = \frac{R_1}{w}\right) \ldots\ldots\ldots\cdots = \frac{R_1^2}{2w}$$

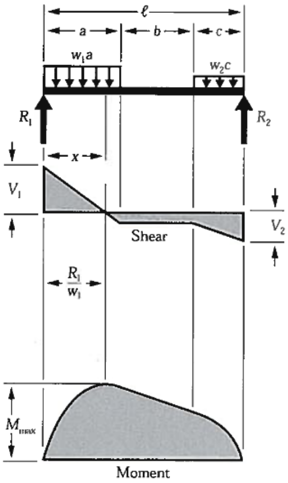
$$M_x \text{ (when } x < a) \ldots\ldots\ldots\cdots = R_1 x - \frac{wx^2}{2}$$

$$M_x \text{ (when } x > a) \ldots\ldots\ldots\cdots = R_2(\ell - x)$$

$$\Delta_x \text{ (when } x < a) \ldots\ldots\ldots\cdots = \frac{wx}{24EI\ell}\left(a^2(2\ell - a)^2 - 2ax^2(2\ell - a) + \ell x^3\right)$$

$$\Delta_x \text{ (when } x > a) \ldots\ldots\ldots\cdots = \frac{wa^2(\ell - x)}{24EI\ell}(4x\ell - 2x^2 - a^2)$$

Figure 7: Case 3



$$R_1 = V_1 \ldots\ldots\ldots\ldots\ldots\cdots = \frac{w_1 a(2\ell - a) + w_2 c^2}{2\ell}$$

$$R_2 = V_2 \ldots\ldots\ldots\ldots\ldots\cdots = \frac{w_2 c(2\ell - c) + w_1 a^2}{2\ell}$$

$$V_x \text{ (when } x < a) \ldots\ldots\ldots\ldots\ldots\cdots = R_1 - w_1 x$$

$$V_x \text{ (when } x > a \text{ and } < (a+b)) \ldots\cdots = R_1 - w_1 a$$

$$V_x \text{ (when } x > (a+b)) \ldots\ldots\ldots\ldots\cdots = R_2 - w_2(\ell - x)$$

$$M_{\max}\left(\text{at } x = \frac{R_1}{w_1} \text{ when } R_1 < w_1 a\right) \ldots\cdots = \frac{R_1^2}{2w_1}$$

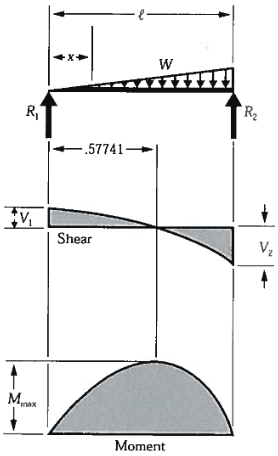
$$M_{\max}\left(\text{at } x = \ell - \frac{R_2}{w_2} \text{ when } R_2 < w_2 c\right) = \frac{R_2^2}{2w_2}$$

$$M_x \text{ (when } x < a) \ldots\ldots\ldots\ldots\ldots\cdots = R_1 x - \frac{w_1 x^2}{2}$$

$$M_x \text{ (when } x > a \text{ and } < (a+b)) \ldots\cdots = R_1 x - \frac{w_1 a}{2}(2x - a)$$

$$M_x \text{ (when } x > (a+b)) \ldots\ldots\ldots\ldots\cdots = R_2(\ell - x) - \frac{w_2(\ell - x)^2}{2}$$

Figure 8: Case 4



$$R_1 = V_1 \ldots\ldots\ldots\ldots\ldots\cdots = \frac{W}{3}$$

$$R_2 = V_2 \ldots\ldots\ldots\ldots\ldots\cdots = \frac{2W}{3}$$

$$V_x \ldots\ldots\ldots\ldots\ldots\cdots = \frac{W}{3} - \frac{Wx^2}{\ell^2}$$

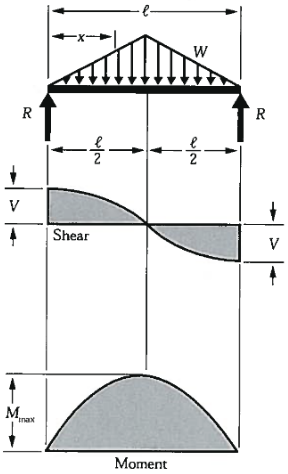
$$M_{\max}\left(\text{at } x = \frac{\ell}{\sqrt{3}} = .5774\ell\right) \ldots\cdots = \frac{2W\ell}{9\sqrt{3}} = .1283W\ell$$

$$M_x \ldots\ldots\ldots\ldots\ldots\cdots = \frac{Wx}{3\ell^2}(\ell^2 - x^2)$$

$$\Delta_{\max}\left(\text{at } x = \ell\sqrt{1 - \sqrt{\frac{8}{15}}} = .5193\ell\right) \cdots = .01304\frac{W\ell^3}{EI}$$

$$\Delta_x \ldots\ldots\ldots\ldots\ldots\cdots = \frac{Wx}{180EI\ell^2}(3x^4 - 10\ell^2 x^2 + 7\ell^4)$$

Figure 9: Case 5



$$R = V \ldots\ldots\ldots\ldots\ldots\cdots = \frac{W}{2}$$

$$V_x \left(\text{when } x < \frac{\ell}{2}\right) \ldots\ldots\ldots\ldots\cdots = \frac{W}{2\ell^2}(\ell^2 - 4x^2)$$

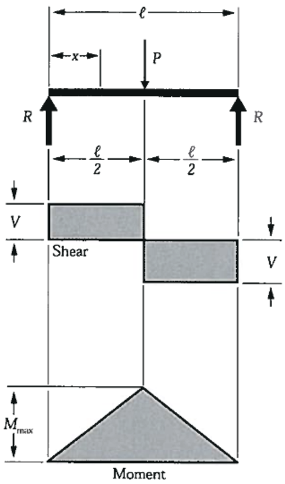
$$M_{\text{max}} \text{ (at center)} \ldots\ldots\ldots\ldots\ldots\cdots = \frac{W\ell}{6}$$

$$M_x \left(\text{when } x < \frac{\ell}{2}\right) \ldots\ldots\ldots\ldots\cdots = Wx\left(\frac{1}{2} - \frac{2x^2}{3\ell^2}\right)$$

$$\Delta_{\text{max}} \text{ (at center)} \ldots\ldots\ldots\ldots\ldots\cdots = \frac{W\ell^3}{60EI}$$

$$\Delta_x \ldots\ldots\ldots\ldots\ldots\cdots = \frac{Wx}{480EI\ell^2}(5\ell^2 - 4x^2)^2$$

Figure 10: Case 6



$$R = V \ldots\ldots\ldots\ldots\ldots\cdots = \frac{P}{2}$$

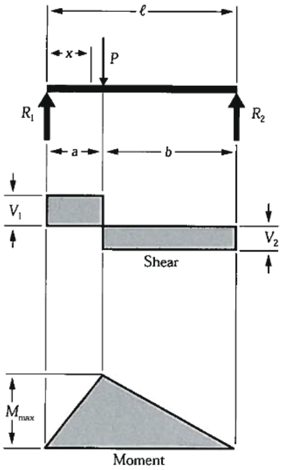
$$M_{\text{max}} \text{ (at point of load)} \ldots\ldots\ldots\ldots\cdots = \frac{P\ell}{4}$$

$$M_x \left(\text{when } x < \frac{\ell}{2}\right) \ldots\ldots\ldots\ldots\cdots = \frac{Px}{2}$$

$$\Delta_{\text{max}} \text{ (at point of load)} \ldots\ldots\ldots\ldots\cdots = \frac{P\ell^3}{48EI}$$

$$\Delta_x \left(\text{when } x < \frac{\ell}{2}\right) \ldots\ldots\ldots\ldots\cdots = \frac{Px}{48EI}(3\ell^2 - 4x^2)$$

Figure 11: Case 7



$$R_1 = V_1 \text{ (max when } a < b) \ldots\ldots\cdots = \frac{Pb}{\ell}$$

$$R_2 = V_2 \text{ (max when } a > b) \ldots\ldots\cdots = \frac{Pa}{\ell}$$

$$M_{\max} \text{ (at point of load)} \ldots\ldots\cdots = \frac{Pab}{\ell}$$

$$M_x \text{ (when } x < b) \ldots\ldots\ldots\cdots = \frac{Pbx}{\ell}$$

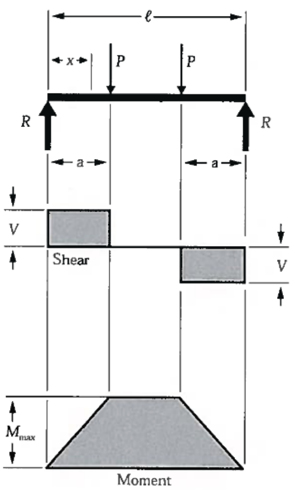
$$\Delta_{\max} \left(\text{at } x = \sqrt{\frac{a(a+2b)}{3}} \text{ when } a > b \right) \cdots = \frac{Pab(a+2b)\sqrt{3a(a+2b)}}{27EI\ell}$$

$$\Delta_a \text{ (at point of load)} \ldots\ldots\cdots = \frac{Pa^2b^2}{3EI\ell}$$

$$\Delta_x \text{ (when } x < a) \ldots\ldots\ldots\cdots = \frac{Pbx}{6EI\ell}(\ell^2 - b^2 - x^2)$$

$$\Delta_x \text{ (when } x > a) \ldots\ldots\ldots\cdots = \frac{Pa(\ell - x)}{6EI\ell}(2\ell x - x^2 - a^2)$$

Figure 12: Case 8



$$R = V \ldots\ldots\ldots\ldots\cdots = P$$

$$M_{\max} \text{ (between loads)} \ldots\ldots\ldots\cdots = Pa$$

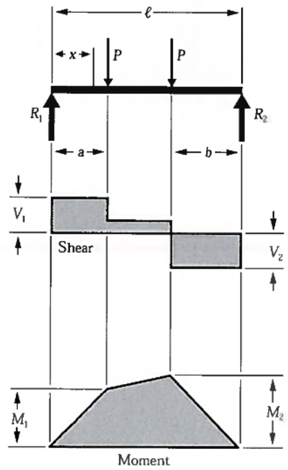
$$M_x \text{ (when } x < a) \ldots\ldots\ldots\cdots = Px$$

$$\Delta_{\max} \text{ (at center)} \ldots\ldots\ldots\cdots = \frac{Pa}{24EI}(3\ell^2 - 4a^2)$$

$$\Delta_x \text{ (when } x < a) \ldots\ldots\ldots\cdots = \frac{Px}{6EI}(3\ell a - 3a^2 - x^2)$$

$$\Delta_x \text{ (when } x > a \text{ and } < (\ell - a)) \ldots\cdots = \frac{Pa}{6EI}(3\ell x - 3x^2 - a^2)$$

Figure 13: Case 9



$$R_1 = V_1 \text{ (max when } a < b) \ldots\ldots\ldots = \frac{P}{\ell}(\ell - a + b)$$

$$R_2 = V_2 \text{ (max when } a > b) \ldots\ldots\ldots = \frac{P}{\ell}(\ell - b + a)$$

$$V_x \text{ (when } x > a \text{ and } < (\ell - b)) \ldots\ldots = \frac{P}{\ell}(b - a)$$

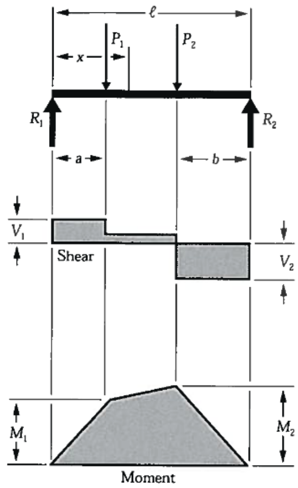
$$M_1 \text{ (max when } a > b) \ldots\ldots\ldots\ldots\ldots = R_1 a$$

$$M_2 \text{ (max when } a < b) \ldots\ldots\ldots\ldots\ldots = R_2 b$$

$$M_x \text{ (when } x < a) \ldots\ldots\ldots\ldots\ldots = R_1 x$$

$$M_x \text{ (when } x > a \text{ and } < (\ell - b)) \ldots\ldots = R_1 x - P(x - a)$$

Figure 14: Case 10



$$R_1 = V_1 \ldots\ldots\ldots\ldots\ldots\ldots\ldots = \frac{P_1(\ell - a) + P_2 b}{\ell}$$

$$R_2 = V_2 \ldots\ldots\ldots\ldots\ldots\ldots\ldots = \frac{P_1 a + P_2(\ell - b)}{\ell}$$

$$V_x \text{ (when } x > a \text{ and } < (\ell - b)) \ldots\ldots = R_1 - P_1$$

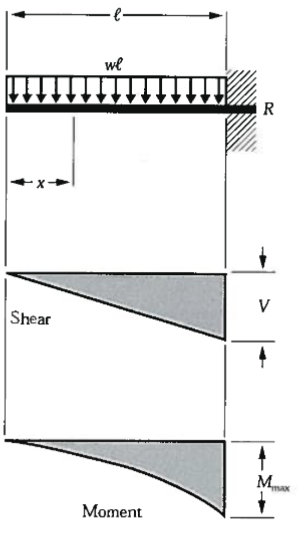
$$M_1 \text{ (max when } R_1 < P_1) \ldots\ldots\ldots\ldots\ldots = R_1 a$$

$$M_2 \text{ (max when } R_2 < P_2) \ldots\ldots\ldots\ldots\ldots = R_2 b$$

$$M_x \text{ (when } x < a) \ldots\ldots\ldots\ldots\ldots = R_1 x$$

$$M_x \text{ (when } x > a \text{ and } < (\ell - b)) \ldots\ldots = R_1 x - P_1(x - a)$$

Figure 15: Case 11



$$R = V \ldots\ldots\ldots\ldots\ldots\ldots\ldots = w\ell$$

$$V_x \ldots\ldots\ldots\ldots\ldots\ldots\ldots = wx$$

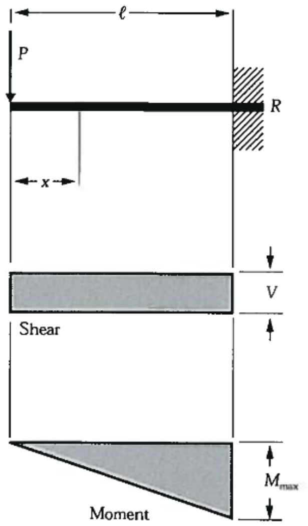
$$M_{\max} \text{ (at fixed end)} \ldots\ldots\ldots\ldots\ldots = \frac{w\ell^2}{2}$$

$$M_x \ldots\ldots\ldots\ldots\ldots\ldots\ldots = \frac{wx^2}{2}$$

$$\Delta_{\max} \text{ (at free end)} \ldots\ldots\ldots\ldots\ldots = \frac{w\ell^4}{8EI}$$

$$\Delta_x \ldots\ldots\ldots\ldots\ldots\ldots\ldots = \frac{w}{24EI}(x^4 - 4\ell^3 x + 3\ell^4)$$

Figure 16: Case 12



$$R = V \dots\dots\dots\dots\dots\dots = P$$

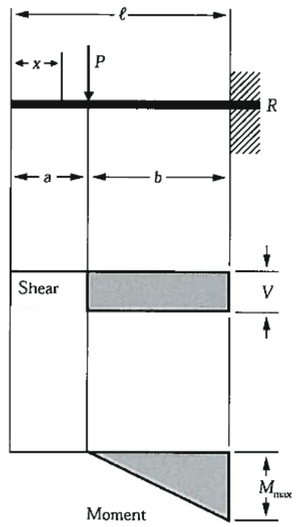
$$M_{\max} \text{ (at fixed end)} \dots\dots\dots\dots = P\ell$$

$$M_x \dots\dots\dots\dots\dots\dots = Px$$

$$\Delta_{\max} \text{ (at free end)} \dots\dots\dots\dots = \frac{P\ell^3}{3EI}$$

$$\Delta_x \dots\dots\dots\dots\dots\dots = \frac{P}{6EI}(2\ell^3 - 3\ell^2 x + x^3)$$

Figure 17: Case 13



$$R = V \dots\dots\dots\dots\dots\dots = P$$

$$M_{\max} \text{ (at fixed end)} \dots\dots\dots\dots = Pb$$

$$M_x \text{ (when } x > a) \dots\dots\dots\dots = P(x - a)$$

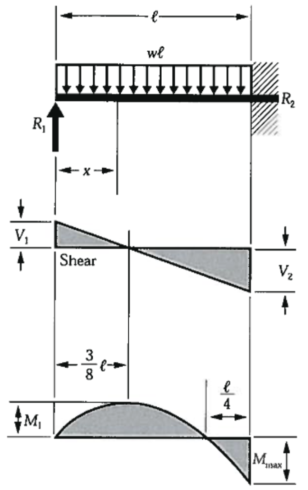
$$\Delta_{\max} \text{ (at free end)} \dots\dots\dots\dots = \frac{Pb^2}{6EI}(3\ell - b)$$

$$\Delta_a \text{ (at point of load)} \dots\dots\dots\dots = \frac{Pb^3}{3EI}$$

$$\Delta_x \text{ (when } x < a) \dots\dots\dots\dots = \frac{Pb^2}{6EI}(3\ell - 3x - b)$$

$$\Delta_x \text{ (when } x > a) \dots\dots\dots\dots = \frac{P(\ell - x)^2}{6EI}(3b - \ell + x)$$

Figure 18: Case 14



$$R_1 = V_1 \dots\dots\dots\dots\dots\dots = \frac{3w\ell}{8}$$

$$R_2 = V_2 \dots\dots\dots\dots\dots\dots = \frac{5w\ell}{8}$$

$$V_x \dots\dots\dots\dots\dots\dots = R_1 - wx$$

$$M_{\max} \dots\dots\dots\dots\dots\dots = \frac{w\ell^2}{8}$$

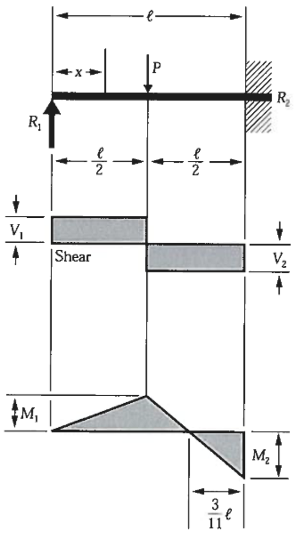
$$M_1 \left(\text{at } x = \frac{3}{8}\ell\right) \dots\dots\dots\dots = \frac{9}{128}w\ell^2$$

$$M_x \dots\dots\dots\dots\dots\dots = R_1 x - \frac{wx^2}{2}$$

$$\Delta_{\max} \left(\text{at } x = \frac{\ell}{16}(1 + \sqrt{33}) = .4215\ell\right) \dots = \frac{w\ell^4}{185EI}$$

$$\Delta_x \dots\dots\dots\dots\dots\dots = \frac{wx}{48EI}(\ell^3 - 3\ell x^2 + 2x^3)$$

Figure 19: Case 15



$$R_1 = V_1 \dots\dots\dots\dots\dots\dots = \frac{5P}{16}$$

$$R_2 = V_2 \dots\dots\dots\dots\dots\dots = \frac{11P}{16}$$

$$M_{\max} \text{ (at fixed end)} \dots\dots\dots\dots = \frac{3P\ell}{16}$$

$$M_1 \text{ (at point of load)} \dots\dots\dots\dots = \frac{5P\ell}{32}$$

$$M_x \left(\text{when } x < \frac{\ell}{2}\right) \dots\dots\dots\dots = \frac{5Px}{16}$$

$$M_x \left(\text{when } x > \frac{\ell}{2}\right) \dots\dots\dots\dots = P\left(\frac{\ell}{2} - \frac{11x}{16}\right)$$

$$\Delta_{\max} \left(\text{at } x = \ell\sqrt{\frac{1}{5}} = .4472\ell\right) \cdots = \frac{P\ell^3}{48EI\sqrt{5}} = .009317\frac{P\ell^3}{EI}$$

$$\Delta_x \text{ (at point of load)} \dots\dots\dots\dots = \frac{7P\ell^3}{768EI}$$

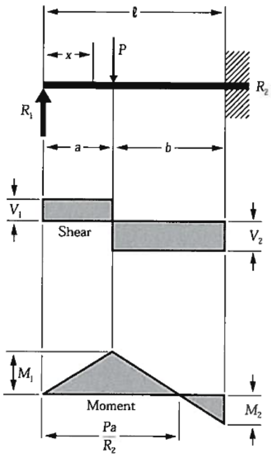
$$\Delta_x \left(\text{when } x < \frac{\ell}{2}\right) \dots\dots\dots\dots = \frac{Px}{96EI}(3\ell^2 - 5x^2)$$

$$\Delta_x \left(\text{when } x > \frac{\ell}{2}\right) \dots\dots\dots\dots = \frac{P}{96EI}(x - \ell)^2(11x - 2\ell)$$

Figure 20: Case 16

Table 7: Symbols and definitions used in core solver functions

Symbol	**Description**	**Unit**
x	Evaluation coordinate along the global span	m
Δx	Finite difference step increment (`dx`)	m
$\mathcal{L}$	Total continuous physical length of the structural profile	m
$y_{\max}$	Distance from neutral axis to the outermost extreme fiber	m
$\Delta(x)$	Elastic beam deflection line profile (downward displacement)	m
$V(x)$	Internal shear force distribution profile field	N
$M(x)$	Internal bending moment distribution profile field	$\mathrm{N \cdot m}$
$\theta(x)$	Elastic slope curve profile (beam rotation field)	rad
$\kappa(x)$	Flexural curvature profile field	m^{-1}
$\sigma_b(x)$	Maximum normal bending stress profile field	Pa
$v_{\max}$	Maximum numerical displacement detected along the span	m
$M_{\max}$	Maximum internal bending moment detected along the span	$\mathrm{N \cdot m}$
$V_{\max}$	Maximum internal shear force magnitude detected along the span	N



$$R_1 = V_1 \dots\dots\dots = \frac{Pb^2}{2\ell^3}(a+2\ell)$$

$$R_2 = V_2 \dots\dots\dots = \frac{Pa}{2\ell^3}(3\ell^2 - a^2)$$

$$M_1 \text{ (at point of load)} \dots\dots\dots = R_1 a$$

$$M_2 \text{ (at fixed end)} \dots\dots\dots = \frac{Pab}{2\ell^2}(a+\ell)$$

$$M_x \text{ (when } x < a) \dots\dots\dots = R_1 x$$

$$M_x \text{ (when } x > a) \dots\dots\dots = R_1 x - P(x-a)$$

$$\Delta_{\max}\left(\text{when } a < .414\ell \text{ at } x = \ell\frac{\ell^2 + a^2}{3\ell^2 - a^2}\right) = \frac{Pa}{3EI}\frac{(\ell^2 - a^2)^3}{(3\ell^2 - a^2)^2}$$

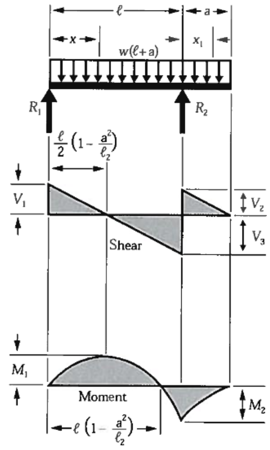
$$\Delta_{\max}\left(\text{when } a > .414\ell \text{ at } x = \ell\sqrt{\frac{a}{2\ell + a}}\right) = \frac{Pab^2}{6EI}\sqrt{\frac{a}{2\ell + a}}$$

$$\Delta_a \text{ (at point of load)} \dots\dots\dots = \frac{Pa^2b^3}{12EI\ell^3}(3\ell + a)$$

$$\Delta_x \text{ (when } x < a) \dots\dots\dots = \frac{Pb^2x}{12EI\ell^3}(3a\ell^2 - 2\ell x^2 - ax^2)$$

$$\Delta_x \text{ (when } x > a) \dots\dots\dots = \frac{Pa}{12EI\ell^3}(\ell - x)^2(3\ell^2 x - a^2 x - 2a^2\ell)$$

Figure 21: Case 17



$$R_1 = V_1 \dots\dots\dots\dots\dots\dots = \frac{w}{2\ell}(\ell^2 - a^2)$$

$$R_2 = V_2 + V_3 \dots\dots\dots\dots\dots\dots = \frac{w}{2\ell}(\ell + a)^2$$

$$V_2 \dots\dots\dots\dots\dots\dots = wa$$

$$V_3 \dots\dots\dots\dots\dots\dots = \frac{w}{2\ell}(\ell^2 + a^2)$$

$$V_x \text{ (between supports)} \dots\dots\dots\dots = R_1 - wx$$

$$V_{x_1} \text{ (for overhang)} \dots\dots\dots\dots = w(a - x_1)$$

$$M_1 \left(\text{at } x = \frac{\ell}{2}\left[1 - \frac{a^2}{\ell^2}\right]\right) \dots\dots\dots = \frac{w}{8\ell^2}(\ell + a)^2(\ell - a)^2$$

$$M_2 \text{ (at } R_2) \dots\dots\dots\dots\dots\dots = \frac{wa^2}{2}$$

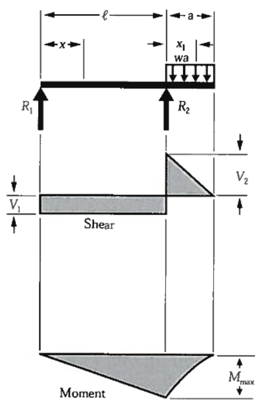
$$M_x \text{ (between supports)} \dots\dots\dots\dots = \frac{wx}{2\ell}(\ell^2 - a^2 - x\ell)$$

$$M_{x_1} \text{ (for overhang)} \dots\dots\dots\dots = \frac{w}{2}(a - x_1)^2$$

$$\Delta_x \text{ (between supports)} \dots\dots\dots\dots = \frac{wx}{24EI\ell}(\ell^4 - 2\ell^2x^2 + \ell x^3 - 2a^2\ell^2 + 2a^2x^2)$$

$$\Delta_{x_1} \text{ (for overhang)} \dots\dots\dots\dots = \frac{wx_1}{24EI}(4a^2\ell - \ell^3 + 6a^2x_1 - 4ax_1^2 + x_1^3)$$

Figure 22: Case 18



$$R_1 = V_1 \dots\dots\dots\dots\dots\dots = \frac{wa^2}{2\ell}$$

$$R_2 = V_1 + V_2 \dots\dots\dots\dots\dots\dots = \frac{wa}{2\ell}(2\ell + a)$$

$$V_2 \dots\dots\dots\dots\dots\dots = wa$$

$$V_{x_1} \text{ (for overhang)} \dots\dots\dots\dots = w(a - x_1)$$

$$M_{\max} \text{ (at } R_2) \dots\dots\dots\dots\dots\dots = \frac{wa^2}{2}$$

$$M_x \text{ (between supports)} \dots\dots\dots\dots = \frac{wa^2x}{2\ell}$$

$$M_{x_1} \text{ (for overhang)} \dots\dots\dots\dots = \frac{w}{2}(a - x_1)^2$$

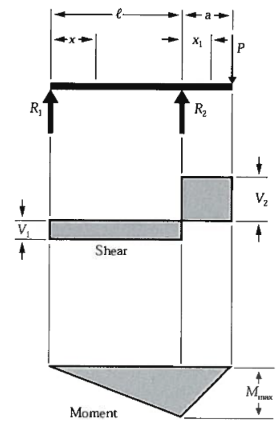
$$\Delta_{\max} \left(\text{between supports at } x = \frac{\ell}{\sqrt{3}}\right) = \frac{wa^2\ell^2}{18\sqrt{3}EI} = .03208\frac{wa^2\ell^2}{EI}$$

$$\Delta_{\max} \text{ (for overhang at } x_1 = a) \dots\dots = \frac{wa^3}{24EI}(4\ell + 3a)$$

$$\Delta_x \text{ (between supports)} \dots\dots\dots\dots = \frac{wa^2x}{12EI\ell}(\ell^2 - x^2)$$

$$\Delta_{x_1} \text{ (for overhang)} \dots\dots\dots\dots = \frac{wx_1}{24EI}(4a^2\ell + 6a^2x_1 - 4ax_1^2 + x_1^3)$$

Figure 23: Case 19



$$R_1 = V_1 \dots\dots\dots\dots = \frac{Pa}{\ell}$$

$$R_2 = V_1 + V_2 \dots\dots\dots\dots = \frac{P}{\ell}(\ell + a)$$

$$V_2 \dots\dots\dots\dots = P$$

$$M_{\max} \text{ (at } R_2) \dots\dots\dots\dots = Pa$$

$$M_x \text{ (between supports)} \dots\dots\dots = \frac{Pax}{\ell}$$

$$M_{x_1} \text{ (for overhang)} \dots\dots\dots = P(a - x_1)$$

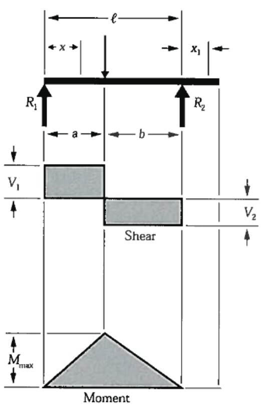
$$\Delta_{\max}\left(\text{between supports at } x = \frac{\ell}{\sqrt{3}}\right) = \frac{Pa\ell^2}{9\sqrt{3}EI} = .06415\frac{Pa\ell^2}{EI}$$

$$\Delta_{\max} \text{ (for overhang at } x_1 = a) \dots\dots = \frac{Pa^2}{3EI}(\ell + a)$$

$$\Delta_x \text{ (between supports)} \dots\dots\dots = \frac{Pax}{6EI\ell}(\ell^2 - x^2)$$

$$\Delta_{x_1} \text{ (for overhang)} \dots\dots\dots = \frac{Px_1}{6EI}(2a\ell + 3ax_1 - x_1^2)$$

Figure 24: Case 20



$$R_1 = V_1 \text{ (max when } a < b) \dots\dots\dots = \frac{Pb}{\ell}$$

$$R_2 = V_2 \text{ (max when } a > b) \dots\dots\dots = \frac{Pa}{\ell}$$

$$M_{\max} \text{ (at point of load)} \dots\dots\dots = \frac{Pab}{\ell}$$

$$M_x \text{ (when } x < a) \dots\dots\dots\dots = \frac{Pbx}{\ell}$$

$$\Delta_{\max}\left(\text{at } x = \sqrt{\frac{a(a+2b)}{3}} \text{ when } a > b\right) \dots = \frac{Pab(a+2b)\sqrt{3a(a+2b)}}{27EI\ell}$$

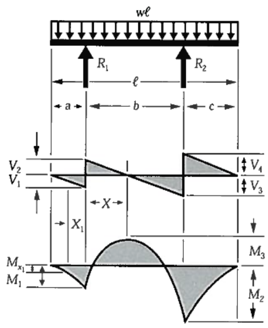
$$\Delta_a \text{ (at point of load)} \dots\dots\dots = \frac{Pa^2b^2}{3EI\ell}$$

$$\Delta_x \text{ (when } x < a) \dots\dots\dots\dots = \frac{Pbx}{6EI\ell}(\ell^2 - b^2 - x^2)$$

$$\Delta_x \text{ (when } x > a) \dots\dots\dots\dots = \frac{Pa(\ell - x)}{6EI\ell}(2\ell x - x^2 - a^2)$$

$$\Delta_{x_1} \dots\dots\dots\dots = \frac{Pabx_1}{6EI\ell}(\ell + a)$$

Figure 25: Case 21



$$R_1 \ldots\ldots\ldots\ldots\ldots = \frac{w\ell(\ell-2c)}{2b}$$

$$R_2 \ldots\ldots\ldots\ldots\ldots = \frac{w\ell(\ell-2a)}{2b}$$

$$V_1 \ldots\ldots\ldots\ldots\ldots = wa$$

$$V_2 \ldots\ldots\ldots\ldots\ldots = R_1 - V_1$$

$$V_3 \ldots\ldots\ldots\ldots\ldots = R_2 - V_4$$

$$V_4 \ldots\ldots\ldots\ldots\ldots = wc$$

$$V_{x_1} \ldots\ldots\ldots\ldots\ldots = V_1 - wx_1$$

$$V_x \text{ (when } x < \ell) \ldots\ldots\ldots = R_1 - w(a + x_1)$$

$$V_m \text{ (when } a < c) \ldots\ldots\ldots = R_2 - wc$$

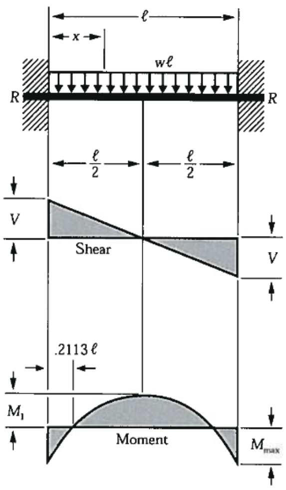
$$M_1 \ldots\ldots\ldots\ldots\ldots = -\frac{wa^2}{2}$$

$$M_2 \ldots\ldots\ldots\ldots\ldots = -\frac{wc^2}{2}$$

$$M_3 \ldots\ldots\ldots\ldots\ldots = R_1\left(\frac{R_1}{2w} - a\right)$$

$$M_x\left(\text{max when } x = \frac{R_1}{w} - a\right) \ldots\ldots = R_1 x - \frac{w(a+x)^2}{2}$$

Figure 26: Case 22



$$R = V \ldots\ldots\ldots\ldots\ldots = \frac{w\ell}{2}$$

$$V_x \ldots\ldots\ldots\ldots\ldots = w\left(\frac{\ell}{2} - x\right)$$

$$M_{\max} \text{ (at ends)} \ldots\ldots\ldots\ldots\ldots = \frac{w\ell^2}{12}$$

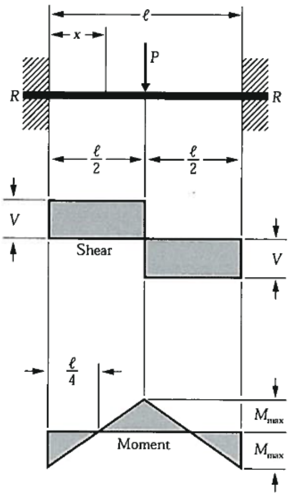
$$M_1 \text{ (at center)} \ldots\ldots\ldots\ldots\ldots = \frac{w\ell^2}{24}$$

$$M_x \ldots\ldots\ldots\ldots\ldots = \frac{w}{12}(6\ell x - \ell^2 - 6x^2)$$

$$\Delta_{\max} \text{ (at center)} \ldots\ldots\ldots\ldots\ldots = \frac{w\ell^4}{3884EI}$$

$$\Delta_x \ldots\ldots\ldots\ldots\ldots = \frac{wx^2}{24EI}(\ell - x)^2$$

Figure 27: Case 23



$$R = V \ldots\ldots\ldots\ldots\ldots\ldots = \frac{P}{2}$$

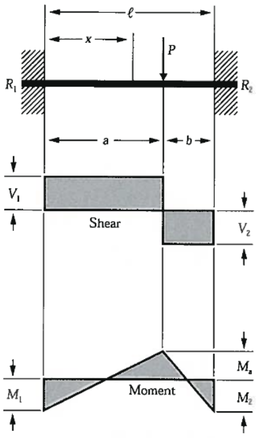
$$M_{\max} \text{ (at center and ends)} \ldots\ldots\ldots\ldots = \frac{P\ell}{8}$$

$$M_x \left(\text{when } x < \frac{\ell}{2}\right) \ldots\ldots\ldots\ldots = \frac{P}{8}(4x - \ell)$$

$$\Delta_{\max} \text{ (at center)} \ldots\ldots\ldots\ldots = \frac{P\ell^3}{192EI}$$

$$\Delta_x \left(\text{when } x < \frac{\ell}{2}\right) \ldots\ldots\ldots\ldots = \frac{Px^2}{48EI}(3\ell - 4x)$$

Figure 28: Case 24



$$R_1 = V_1 \text{ (max when } a < b) \ldots\ldots\ldots = \frac{Pb^2}{\ell^3}(3a + b)$$

$$R_2 = V_2 \text{ (max when } a > b) \ldots\ldots\ldots = \frac{Pa^2}{\ell^3}(a + 3b)$$

$$M_1 \text{ (max when } a < b) \ldots\ldots\ldots\ldots = \frac{Pab^2}{\ell^2}$$

$$M_2 \text{ (max when } a > b) \ldots\ldots\ldots\ldots = \frac{Pa^2b}{\ell^2}$$

$$M_a \text{ (at point of load)} \ldots\ldots\ldots\ldots = \frac{2Pa^2b^2}{\ell^3}$$

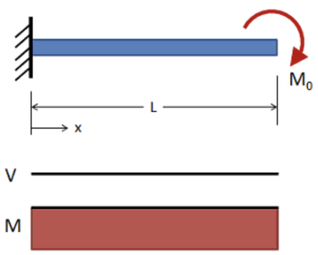
$$M_x \text{ (when } x < a) \ldots\ldots\ldots\ldots = R_1 x - \frac{Pab^2}{\ell^2}$$

$$\Delta_{\max} \left(\text{when } a > b \text{ at } x = \frac{2a\ell}{3a + b}\right) \ldots\ldots = \frac{2Pa^3b^2}{3EI(3a + b)^2}$$

$$\Delta_a \text{ (at point of load)} \ldots\ldots\ldots\ldots = \frac{Pa^3b^3}{3EI\ell^3}$$

$$\Delta_x \text{ (when } x < a) \ldots\ldots\ldots\ldots = \frac{Pb^2x^2}{6EI\ell^3}(3a\ell - 3ax - bx)$$

Figure 29: Case 25



$$R = V \ldots\ldots\ldots\ldots\ldots\ldots = 0$$

$$V_x \ldots\ldots\ldots\ldots\ldots\ldots = 0$$

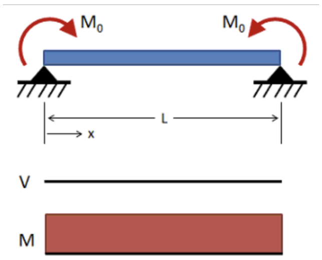
$$M_{\max} \text{ (at fixed end)} \ldots\ldots\ldots\ldots = -M_0$$

$$M_x \ldots\ldots\ldots\ldots\ldots\ldots = -M_0$$

$$\Delta_{\max} \text{ (at free end)} \ldots\ldots\ldots\ldots = \frac{M_0\ell^2}{2EI}$$

$$\Delta_x \ldots\ldots\ldots\ldots\ldots\ldots = -\frac{M_0x^2}{2EI}$$

Figure 30: Case 26: Cantilever, End Moment



$$R = V \ldots\ldots\ldots\ldots\ldots\ldots = 0$$

$$V_x \ldots\ldots\ldots\ldots\ldots\ldots = 0$$

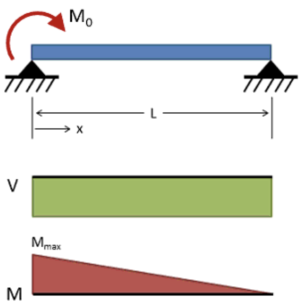
$$M_{\text{max}}\ (\text{at center}) \ldots\ldots\ldots\ldots\ldots\ldots = M_0$$

$$M_x \ldots\ldots\ldots\ldots\ldots\ldots = M_0$$

$$\Delta_{\text{max}} \left(\text{at } x = \frac{\ell}{2}\right) \ldots\ldots\ldots\ldots = \frac{M_0 \ell^2}{8EI}$$

$$\Delta_x \ldots\ldots\ldots\ldots\ldots\ldots = -\frac{M_0 x}{2EI}(\ell - x)$$

Figure 31: Case 27: Simply Supported, Moment at Each Support



$$R = V \ldots\ldots\ldots\ldots\ldots\ldots = -\frac{M_0}{\ell}$$

$$V_x \ldots\ldots\ldots\ldots\ldots\ldots = -\frac{M_0}{\ell}$$

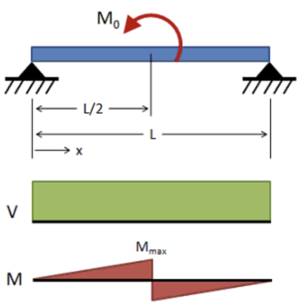
$$M_{\text{max}}\ (\text{at } x = 0) \ldots\ldots\ldots\ldots\ldots\ldots = M_0$$

$$M_x \ldots\ldots\ldots\ldots\ldots\ldots = M_0 \left(1 - \frac{x}{\ell}\right)$$

$$\Delta_{\text{max}} \left(\text{at } x = \ell \left(1 - \frac{\sqrt{3}}{3}\right)\right) \cdots = \frac{M_0 \ell^2}{9\sqrt{3}EI}$$

$$\Delta_x \ldots\ldots\ldots\ldots\ldots\ldots = -\frac{M_0 x}{6\ell EI}(2\ell^2 - 3\ell x + x^2)$$

Figure 32: Case 28: Simply Supported, Moment at One Support



$$R = V \ldots\ldots\ldots\ldots\ldots\ldots = \frac{M_0}{\ell}$$

$$V_x \ldots\ldots\ldots\ldots\ldots\ldots = \frac{M_0}{\ell}$$

$$M_{\text{max}} \left(\text{at } x = \frac{\ell}{2}\right) \ldots\ldots\ldots\ldots\ldots = \frac{M_0}{2}$$

$$M_x \left(\text{when } x < \frac{\ell}{2}\right) \ldots\ldots\ldots\ldots = \frac{M_0 x}{\ell}$$

$$\Delta_x \left(\text{when } x < \frac{\ell}{2}\right) \ldots\ldots\ldots\ldots = -\frac{M_0 x}{24\ell EI}(\ell^2 - 4x^2)$$

Figure 33: Case 29: Simply Supported, Center Moment